

# Vishal Bhatoy

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## EDUCATION

### Carleton University

PhD in Pure Mathematics

Advisor: Dr. Colin Ingalls

Dissertation: *4-dimensional Artin-Schelter regular algebras and invariants via superpotentials*

Ottawa, ON

Jan 2021 – Aug 2025

### Guru Nanak Dev University

M.Sc. Mathematics (CGPA: 9.42/10, Rank-I in University)

B.Sc. (Mathematics, Physics, Chemistry) (CGPA: 8/10, Distinction)

Amritsar, India

July 2017 – June 2019

July 2014 – June 2017

## Research Interests

Algebraic Geometry, Mathematical Physics, Calabi–Yau manifolds, Hilbert schemes of points on surfaces, Homological invariants.

## RESEARCH EXPERIENCE

### Carleton University

*Research Associate*

*Research Assistant*

Ottawa, ON

Sept 2025 – present

Jan 2021 – Aug 2025

- Research in algebraic geometry using computer algebra systems (Magma, Macaulay2, GAP). Studied Koszul Artin-Schelter regular algebras via superpotentials and tensor decompositions. Computed Hochschild cohomology and Kodaira-Spencer maps for 77 known families, identifying 45 components of the moduli stack. Calculated discrete invariants to distinguish moduli components.

### Simons Laufer Mathematical Sciences Institute (SLMath)

*Program Associate (Noncommutative Algebraic Geometry)*

Berkeley, CA

Jan 2024 – May 2024

Presented a team poster and participated in seminars and workshops. Collaborated with mathematicians on group research projects. Participated in office hours and discussions on open problems.

## PUBLICATIONS

1. **Discrete Invariants of Koszul Artin-Schelter Regular Algebras of Dimension Four.** arXiv:2602.13178  
**Authors:** Vishal Bhatoy and Colin Ingalls  
**Submitted to:** *Journal of Algebras and Representation Theory*
2. **Some Components of the Moduli Space of Koszul Artin-Schelter Regular Algebras of Dimension Four.** arXiv:2511.08390  
**Authors:** Vishal Bhatoy, Colin Ingalls, Félix Laroche, and Ravali Nookala  
**Submitted to:** *Journal of Algebra*

## WORKSHOPS / CONFERENCES

**Carleton University** Ottawa, ON  
Joint UOttawa/Carleton Algebra Seminar Feb 27, 2026

**Presented a talk on “Log Calabi-Yau algebras”**

**University of Washington** Seattle, WA  
Noncommutative Algebra Conference Dec 15–19, 2025

**University of California** Berkeley, CA  
Infinite-Dimensional Division Algebras Conference at SLMath Dec 1–5, 2025

**University of Sherbrooke** Sherbrooke, QC  
35th Meeting on the Representation Theory of Algebras and Related Topics Oct 23–25, 2025

**Queen’s University** Kingston, ON  
2025 Route 81 Mathematics Conference Sept 26–27, 2025

**Carleton University** Ottawa, ON  
Joint UOttawa/Carleton Algebra Seminar Jan 31, 2025

**Presented a talk on “Discrete Invariants of Artin-Schelter regular algebras”**

**University of Sherbrooke** Sherbrooke, QC  
Representation Theory meeting Oct 4–5, 2024

**Presented a talk on “Some Components of the moduli space of Artin-Schelter regular algebras”**

**University of California** Berkeley, CA  
Conferences and Workshops organised by SLMath Jan 16 – May 24, 2024

**Presented a poster on “Artin-Schelter regular algebras of dimension four”**

## AWARDS AND HONORS

- International Doctoral Tuition Scholarship (IDTS), Carleton University, 2021-2025.
- D Chand and R Devi Marwah Scholarship for outstanding performance as a graduate student, 2024-2025, Carleton University.
- GSA (Graduate Student Association) Scholarship for excellence in graduate research and academic achievements, 2023, 2024 & 2025, Carleton University.
- Gold Medal for excellence as the top graduate (1st position) in the Master’s program, 2019, Guru Nanak Dev University.
- Lala Bindra Ban Sondhi scholarship for academic excellence (1st position) in the Bachelor’s program, 2017, Doaba College.

## EXTRA-CURRICULAR ACTIVITIES

- Organize weekly seminars on Algebra and Algebraic Geometry, Carleton University, Sept 2025–Present.
- Volunteered in educational programs for marginalized and Indigenous communities, McGill University, Oct–Dec 2025.
- Organized and facilitated study sessions and coordinated major events, including preparation for the Kangaroo Math Competition, Carleton University, March 24, 2024.
- Organized the International Conference in conjunction with the 14<sup>th</sup> Biennial Conference of ISIAM, Guru Nanak Dev University, Aug 23, 2017.